

Free¹ Versions of the Strong Szegő Limit Theorem

Mike Jury (me) Vikus van Rensburg George Roman

University of Florida

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¹(“free” as in “freedom”)

the classical SSLT

setup:

- ▶ f - an L^1 function on the unit circle
- ▶ $a_n = \widehat{f}(n)$ - Fourier coefficients
- ▶ form the Toeplitz matrix

$$T_N(f) = \begin{bmatrix} a_0 & a_{-1} & a_{-2} & \cdots & a_{-N} \\ a_1 & a_0 & a_{-1} & \cdots & a_{-(N-1)} \\ a_2 & a_1 & a_0 & \cdots & a_{-(N-2)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_N & a_{N-1} & a_{N-2} & \cdots & a_0 \end{bmatrix} .$$

- ▶ $D_N(f) = \det T_N(f)$ - a Toeplitz determinant

the classical SSLT

Theorem (classical SSLT — determinant form)

If $f = e^\phi$ with Fourier coefficients $\hat{\phi}(n) = c_n$, then we have the asymptotic

$$D_N(e^\phi) \sim E_0^{N+1} \cdot E_1,$$

where

- ▶ $E_0 = \exp(c_0) = \exp \int \log f$
- ▶ $E_1 = \exp \left(\frac{1}{2} \sum_{n=-\infty}^{\infty} c_n c_{-n} |n| \right)$
- ▶ (...history of gradually reduced smoothness assumptions on ϕ from [Szegő 1952] to [Ibragimov 1967]...)

probabilistic reformulation

a magic formula: Via the “Coulomb gas representation” and the Weyl integration formula, if U is a Haar-distributed $N \times N$ unitary matrix

$$D_N(e^\phi) = \mathbb{E}_{N+1}[\exp(\text{Tr } \phi(U))].$$

So we get

Theorem (Classical SSLT - probabilistic form)

$$\mathbb{E}_N[\exp(\text{Tr } \phi(U))] \sim E_0^N \cdot E_1$$

Put $F_N = \exp(\text{Tr } \phi(U))$... a corollary says $F_N - \mathbb{E}F_N$ converges in distribution to a Gaussian (though SSLT is stronger than this)

background/philosophy

[Diaconis-Shahshahani (1994)]: for $N \times N$ random unitary U ,

$$\text{Tr}_N U^k \rightarrow \mathcal{Z}(0, k) \quad (\text{complex Gaussian})$$

in distribution as $N \rightarrow \infty$ (any integer k)

- ▶ Proof idea: representation theory + combinatorics allow explicit moment calculations
- ▶ More generally: $\text{Tr}_N \phi(U)$ should converge in distribution to Gaussians
- ▶ SSLT can be deduced from this + more combo & rep. theory [Bump-Diaconis (2002)]

alternate philosophy: convergence in distribution plus concentration of measure can prove SSLT-shaped theorems

from one unitary to many

- ▶ Consider g independent Haar-distributed unitaries $U_1, \dots, U_g$.
- ▶ The correct framework is **free probability theory** (Voiculescu).
- ▶ Functions on the circle are indexed by $\mathbb{Z}$ (Fourier series).
The noncommutative analogue: index by words in the **free group** $\mathbb{F}_g$.

tracial free power series

For $c : \mathbb{F}_g \rightarrow \mathbb{C}$, form

$$G_N = \text{Tr} \sum_{w \in \mathbb{F}_g} c_w U^w,$$

where U^w is the product of the U_j in the order prescribed by w .

Example: $w = \ell_2 \ell_1 \ell_3 \Rightarrow U^w = U_2 U_1 U_3$.

free probability

notation:

- ▶ $u_1, \dots, u_g$: canonical generators of $\mathbb{F}_g$ in the regular representation (a.k.a. free Haar unitaries)
- ▶ τ - unique normalized tracial state on $C_{red}^*(\mathbb{F}_g)$

Big idea (Voiculescu): the u_i serve as a deterministic “model” for large independent random unitaries U_i :

- ▶ (Voiculescu (1991)) limits of normalized traces:
for any free $*$ -polynomial p ,

$$\lim_{N \rightarrow \infty} \frac{1}{N} \text{Tr} [p(U, U^*)] \rightarrow \tau(p(u, u^*))$$

almost surely

free probability continued

second order freeness [Mingo, Śniady, Speicher (2007)]:

for (cyclically reduced) words, we have limits of covariances of un-normalized traces

$$\blacktriangleright \quad \lim_{N \rightarrow \infty} \mathbb{E}_N [Tr(U^w) \cdot Tr(U^v)] = \varphi_2(u^w, u^v)$$

where

$$\varphi_2(u^w, u^v) = \text{number of rotations of } v \text{ equal to } w^{-1}$$

background/philosophy redux

[Mingo-Śniady-Speicher (2007)]: for $N \times N$ independent random unitary $U_1, \dots, U_g$

$Tr_N U^w \rightarrow \mathcal{Z}$ complex Gaussian, with mean 0 and a certain variance...

in distribution as $N \rightarrow \infty$

- ▶ Proof idea: this is part of the “second order freeness package”
- ▶ More generally: $Tr_N \phi(U)$ should converge in distribution to Gaussians
- ▶ ...can an SSLT be deduced from this?...
(one-variable combinatorial approach of Bump-Diaconis does not apply)

alternate philosophy: second order freeness plus concentration of measure can prove SSLT-shaped theorems

the “easy” free SSLT

Theorem (J. - van Rensburg - Roman 2026+)

Let $c : \mathbb{F}_g \rightarrow \mathbb{C}$ be an admissible² function with $\sum_w |c_w| |w| < \infty$.
Let $\varphi(U) = \sum c_w U^w$. Then as $N \rightarrow \infty$,

$$\mathbb{E}_N[\exp \operatorname{Tr} \varphi(U)] \sim E_0^N \cdot E_1$$

where

► $E_0 = \exp(c_\emptyset)$

► $E_1 = \exp \left(\frac{1}{2} \sum_{w \in \mathbb{F}_g} c_w c_{w^{-1}} |w| \right)$

Remark: conclusion is essentially identical to $g = 1$ case, but **summability hypothesis** is awfully strong

²supported on cyclically reduced words; constant over cyclic rotations

proof ingredients 1: concentration of measure

Theorem (Meckes and Meckes, 2013)

For Lipschitz³ functions $F_N : \mathcal{U}(N)^g \rightarrow \mathbb{R}$, with constant L ,

$$\mathbb{P}(F_N(U) - \mathbb{E}F_N(U) \geq t) \leq \exp\left(-\frac{Nt^2}{6L^2}\right).$$

for all $t > 0$.

³with respect to the Hilbert-Schmidt metric $d(U, V) = (\sum \|U_j - V_j\|_2^2)^{1/2}$

proof ingredients 1: concentration of measure

Corollary (Unifrom sub-Gaussian tails)

Let $F_N : \mathcal{U}(N)^g \rightarrow \mathbb{C}$ satisfy:

- ▶ Lipschitz constants of F_N are $O(\sqrt{N})$, and
- ▶ $\sup_N |\mathbb{E} F_N| < \infty$.

Then there exist $C, c > 0$ such that for all N and all $t > 0$

$$\mathbb{P}(|F_N| > t) \leq C \exp(-ct^2).$$

proof ingredients 2: second order freeness

second order freeness will say that for a finite sum:

$$\begin{aligned} \lim_{N \rightarrow \infty} \mathbb{E} \left[\text{Tr} \left(\sum c_w U^w \right), \text{Tr} \left(\sum c_w U^w \right) \right] \\ = \varphi_2 \left(\sum c_w u^w, \sum c_w u^w \right) \\ = \sum_w c_w c_{w^{-1}} |w| \end{aligned}$$

(provided c_w is supported on cyclically reduced words and invariant under rotation)

proof summary: “easy” free SSLT

Let

$$F_N = \text{Tr} \sum c_w U^w, \quad \text{with } \sum |c_w| |w| < \infty, \quad c_\emptyset = 0$$

suffices to prove $\mathbb{E}[\exp(F_N)] \rightarrow \exp\left(\frac{1}{2} \sum c_w c_{w^{-1}} |w|\right)$

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- ▶ second order freeness says $F_N \rightarrow$ Gaussian in distribution;
limiting variance is $\sum c_w c_{w^{-1}} |w|$

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- ▶ if $C = \sum |c_w| |w| < \infty$ then F_N are Lipschitz with constant $C\sqrt{N}$

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- concentration of measure implies uniform integrability of the F_N

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- uniform integrability upgrades convergence in distribution to convergence of $\mathbb{E}[\exp(F_N)]$

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► finish by approximation argument

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- ▶ first assume a finite sum
- ▶ second order freeness says $F_N \rightarrow$ Gaussian in distribution; limiting variance is $\sum c_w c_{w^{-1}} |w|$
- ▶ if $C = \sum |c_w| |w| < \infty$ then F_N are Lipschitz with constant $C\sqrt{N}$
- ▶ concentration of measure implies uniform integrability of the F_N
- ▶ uniform integrability upgrades convergence in distribution to convergence of $\mathbb{E}[\exp(F_N)]$
- ▶ finish by approximation argument

beyond the “easy” theorem: a special case

Given $k \times k$ coefficient matrices $A_1, \dots, A_g$, the **monic linear pencil** is

$$L_A(U^{(N)}) = I_k \otimes I_N + \sum_{j=1}^g A_j \otimes U_j^{(N)},$$

a $(kN) \times (kN)$ random matrix.

One-Variable Benchmark

The classical SSLT gives, for $\rho(A) < 1, \rho(B) < 1$:

$$\lim_{N \rightarrow \infty} \mathbb{E}_N \left[\det(I + A \otimes U) \overline{\det(I + B \otimes U)} \right] = \det(I - A \otimes \overline{B})^{-1}.$$

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proof idea:

- ▶ $\mathbb{E}[\det \varphi(U)] = \mathbb{E}[\exp \operatorname{Tr}(\log \varphi(U))]$

- ▶ expand

$$\operatorname{Tr} [\log(I + (\text{contractive stuff}))]$$

as convergent series in $\operatorname{Tr}(U^n)$

- ▶ apply classical SSLT (no problem)

beyond the “easy” theorem: a special case

definition: the **outer spectral radius**⁴:

$$\rho_{out}(A) = \rho \left(\sum_{j=1}^g \overline{A_j} \otimes A_j \right)^{1/2},$$

Theorem (J.-van Rensburg-Roman 2026+)

Let $A_1, \dots, A_g$ ($k \times k$) and $B_1, \dots, B_g$ ($\ell \times \ell$) satisfy $\rho_{out}(A), \rho_{out}(B) < 1$. Then

$$\lim_{N \rightarrow \infty} \mathbb{E}_N \left[\det L_A(U^{(N)}) \overline{\det L_B(U^{(N)})} \right] = \det \left(I_{k\ell} - \sum_{j=1}^g A_j \otimes \overline{B_j} \right)^{-1}.$$

⁴Bunce 1984, Popescu 2014, Pascoe 2019, Shalit-Shamovich 2025

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- special cases proved in [J.-Roman 2026] by combinatorial methods

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- BIG DIFFERENCE: this does NOT reduce to an instance of the free SSLT (unlike 1-variable)

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► so why the outer spectral radius ρ_{out} ?

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- ▶ special cases proved in [J.-Roman 2026] by combinatorial methods
- ▶ BIG DIFFERENCE: this does NOT reduce to an instance of the free SSLT (unlike 1-variable)
- ▶ the culprit is the spectral radius hypothesis...we are allowing “bigger” A, B than expected
- ▶ so why the outer spectral radius ρ_{out} ?
- ▶ need one more idea from free probability: strong convergence

why the outer spectral radius?

- ▶ if we had

$$\rho\left(\sum A_j \otimes U_j^{(N)}\right) \leq r < 1$$

for all U , the one variable idea would work

- ▶ BUT in general it is possible that

$$\rho_{out}(A) < 1 \quad \text{while} \quad \rho\left(\sum A_j \otimes U_j^{(N)}\right) > 1$$

- ▶ ...this is worst case behavior, what is typical?

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the free probability heuristic:

- ▶ bulk behavior of $\sum A_j \otimes U_j^{(N)}$, as $N \rightarrow \infty$...

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the free probability heuristic:

- ▶ bulk behavior of $\sum A_j \otimes U_j^{(N)}$, as $N \rightarrow \infty$...
- ▶ ...should be modeled by the free limit $\sum A_j \otimes u_j$

So, the question:

$$\text{What is } \rho\left(\sum A_j \otimes u_j\right) ?$$

spectral radius of free Haar pencils

Lemma (J.-van Rensburg-Roman 2026+)

For matrices $X_1, \dots, X_g$

$$\rho\left(\sum_{j=1}^g X_j \otimes u_j\right) = \rho\left(\sum_{j=1}^g \overline{X_j} \otimes X_j\right)^{1/2} := \rho_{out}(X)$$

Proof idea:

Haagerup–Pisier [1993] and Buchholz [1999] inequalities for free groups give adequate upper and lower bounds on

$$\left\| \left(\sum_{j=1}^g X_j \otimes u_j \right)^n \right\| = \left\| \sum_{|w|=n} X^w \otimes u^w \right\|$$

last ingredient: strong convergence and Parraud's estimate

we need an upgrade of Voiculescu's theorem

$$\lim_{N \rightarrow \infty} \frac{1}{N} \text{Tr } p(U, U^*) \rightarrow \tau(p(u, u^*))$$

to allow for smooth functions, not just polynomials...

this is part of the strong convergence story, begun in [Haagerup-Thorbjørnsen (2005)]

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Theorem (Parraud 2022)

For p a selfadjoint polynomial and $f : \mathbb{R} \rightarrow \mathbb{R}$ a C^6 function,

$$\mathbb{E} \frac{1}{N} \text{Tr}(f(p(U, U^*))) \rightarrow \tau(f(p(u, u^*)))$$

at a rate $O(\frac{\log^2 N}{N^2})$

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Key for us:

- ▶ when $\tau(f(p(u, u^*))) = 0$, the averaged UN-normalized traces of $f(p(U, U^*))$ stay bounded (in fact $\rightarrow 0$)

proof idea: free SSLT for linear pencils

for small enough A , “easy” theorem applies, extend to bigger A by analytic continuation:

start with $F_N = \text{Tr} \log L_A(U) L_A(U)^*$

enough to prove uniform bounds on $\mathbb{E}[\exp F_N]$

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- ▶ F_N is NOT Lipschitz...so introduce a suitable smooth cutoff f of the log function....
...from outer spectral radius business, $L_A(u) L_A(u)^*$ is positive definite
...so choose cutoff to agree with $\log x$ on spectrum of $L_A(u) L_A(u)^*$ (+ a little more)

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- ▶ make new $\widetilde{F}_N$, using cutoff in place of log
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a free⁵ upgrade to the free SSLT for pencils

- call a free polynomial stable if $\det p(X) \neq 0$ for all $\|X\|_{\text{row}} \leq 1$

Theorem (J.-van Rensburg-Roman 2026+)

For stable p, q there exist A, B with $\rho_{\text{out}}(A), \rho_{\text{out}}(B) < 1$ such that $\det p(U) = \det L_A(U)$ etc., and

$$\lim_{N \rightarrow \infty} \mathbb{E}_N[\det p(U) \overline{\det q(U)}] = \det \left(I - \sum A_j \otimes \overline{B_j} \right)^{-1}$$

⁵(“free” as in “beer”)

an open problem: higher asymptotics?

For $F_N = \text{Tr} \sum_w c_w U^w = \text{Tr} \phi(U)$ we have essentially

$$\mathbb{E}[\exp(F_N)] \sim E_0^N E_1$$

or, taking logs,

$$\log \mathbb{E}_N [\exp F_N] = Nc_\emptyset + \frac{1}{2} \sum_{w \in \mathbb{F}_g} c_w c_{w^{-1}} |w| + o(1)$$

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- ▶ one variable, very smooth ϕ : actually $o(1) = O(\exp(-\beta N))$ (convergence to Gaussians is extremely rapid!)
- ▶ in free setting, numerical evidence suggests only $o(1) = O\left(\frac{1}{N}\right)$

Problem: Describe the error term in the free setting. Is there an asymptotic expansion in $\frac{1}{N}$?